

Fourier-algebra restriction to isotropy is always surjective

A full answer to Remark 3.10 of arXiv:2411.14624

Candidate solution packet — run `fa_banach_001`, lane 2

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Status. Candidate full solution, likely valid; human review is recommended. For every locally compact Hausdorff groupoid with a left Haar system, restriction from the groupoid Fourier algebra to the Fourier algebra of any isotropy group is surjective. No countability, étale, unimodularity, or right-invariance hypothesis is required.

1 Source question

Joseph DeGaetani and Mahya Ghandehari, *On the restriction maps of the Fourier and Fourier-Stieltjes algebras over locally compact groupoids*, arXiv:2411.14624, prove the result for étale groupoids in Theorem 3.7. Remark 3.10 (printed page 13) asks whether that theorem is true in full generality. The exact source passage is reproduced below.

So the two representations σ and $L_{\mathcal{G}_u^u}$ are quasi-equivalent (by [18, Proposition 5.3.1]), which implies that $A_\sigma(\mathcal{G}_u^u) = A(\mathcal{G}_u^u)$ (by [31, Proposition 2.8.12]). $\square$

Remark 3.10. The condition that $\mathcal{G}$ is étale is not strictly necessary, though we do not know if Theorem 3.7 is true in full generality. We explicitly use the étale condition for two reasons. Firstly, to make sure that $\mathcal{G}$ satisfies Condition (*) in Definition 3.2; and secondly, to guarantee that the maps U_v are isometric. In the latter consideration, one needs the measure λ^u of the groupoid's Haar system to be both left and right invariant, that is, in addition to Definition 2.3 (3), we need to have

$$\int_{\mathcal{G}^u} f(xz) \lambda^u(x) = \int_{\mathcal{G}^u} f(x) d\lambda^u(x)$$

for $f \in L^2(\mathcal{G}^u)$. This is certainly true for étale groupoids, since the collection of counting measures is the chosen Haar system in this scenario. This condition will also be true for group bundles which have a Haar system for all cases of $u \in \mathcal{G}^{(0)}$ such that $\mathcal{G}^u$ is a unimodular group.

In symbols, if $\mathcal{G}$ is a locally compact Hausdorff groupoid with a left Haar system $\{\lambda^v\}_{v \in \mathcal{G}^{(0)}}$, $u \in \mathcal{G}^{(0)}$, and $H = \mathcal{G}_u^u$, the question is whether

$$R_u : A(\mathcal{G}) \longrightarrow A(H), \quad R_u \varphi = \varphi|_H,$$

is onto without assuming that $\mathcal{G}$ is étale.

2 Definitions and the source reduction

Put $X = \mathcal{G}^u = r^{-1}(u)$ and $\mu = \lambda^u$. The isotropy group $H = \mathcal{G}_u^u$ acts on X by left multiplication. Fix a left Haar measure dh on H . Write λ for the left regular representation on $L^2(H, dh)$,

$$(\lambda(k)\xi)(h) = \xi(k^{-1}h),$$

and let σ be the restriction to H of the groupoid left-regular representation,

$$(\sigma(k)g)(x) = g(k^{-1}x), \quad g \in L^2(X, \mu).$$

For a unitary representation π of H , $A_\pi(H)$ denotes the closed linear span in $B(H)$ of its coefficient functions. Thus $A_\lambda(H) = A(H)$.

Two general steps already contained in the proof of Theorem 3.7 of the source paper do not use etaleness:

$$R_u(A(\mathcal{G})) = A_\sigma(H) \subseteq A(H). \quad (1)$$

Indeed, each coefficient of σ lifts to a coefficient of the groupoid regular representation by the continuous-section extension theorem. An absolutely summable coefficient decomposition in $A_\sigma(H)$ then lifts term by term, proving surjectivity onto $A_\sigma(H)$. Conversely, restriction of compactly supported groupoid coefficients lies in $C_c(H) \cap B(H)$, and continuity gives the inclusion in $A(H)$. The source uses etaleness only in its subsequent attempt to identify $A_\sigma(H)$ with $A(H)$ by splitting X into individual source fibers. We replace that step.

We shall use only the elementary coefficient-norm estimate

$$\|\pi_{\xi,\eta}\|_{B(H)} \leq \|\xi\| \|\eta\|.$$

3 Proof intuition

The tempting proof disintegrates μ over $H \backslash X$ and identifies σ with an amplification of λ . That route introduces second-countability and measurable-selection hypotheses which are absent from the question.

Instead, average a compactly supported function on X along each H -orbit. For $f \in C_c(X)$ this produces a bounded intertwiner $W_f : L^2(H) \rightarrow L^2(X)$. Properness of the free isotropy action gives the bound. More importantly, the whole family $\{W_f : f \in C_c(X)\}$ separates the vectors of $L^2(H)$: restrict to one closed orbit and extend an arbitrary $C_c(H)$ test function to $C_c(X)$. Thus the adjoint ranges $\text{ran } W_f^*$ span $L^2(H)$ densely. A regular coefficient whose two vectors lie in finite sums of these ranges is, by the intertwining equations, a finite sum of σ -coefficients. Approximation in coefficient norm yields $A(H) \subseteq A_\sigma(H)$, and (1) finishes the proof.

4 The theorem

Theorem 1 (Full restriction theorem). *Let $\mathcal{G}$ be a locally compact Hausdorff groupoid admitting a left Haar system. For every $u \in \mathcal{G}^{(0)}$, the restriction map*

$$R_u : A(\mathcal{G}) \longrightarrow A(\mathcal{G}_u^u)$$

is a contractive surjective algebra homomorphism.

The contractive homomorphism assertion is Proposition 3.6 of the source. We prove surjectivity through three lemmas.

Lemma 2 (The isotropy action). *The left action of H on X is free and proper.*

Proof. If $hx = x$, multiplication on the right by x^{-1} gives $h = u$, so the action is free. Consider the shear map

$$\Theta : H \times X \longrightarrow X \times X, \quad \Theta(h, x) = (hx, x).$$

Its image is the equivalence relation

$$\mathcal{R} = \{(y, x) \in X \times X : s(y) = s(x)\}.$$

The inverse on $\mathcal{R}$ is $(y, x) \mapsto (yx^{-1}, x)$, so Θ is a homeomorphism onto $\mathcal{R}$. Since $\mathcal{G}^{(0)}$ is Hausdorff and s is continuous, $\mathcal{R}$ is closed in $X \times X$. Therefore Θ is proper, which is precisely properness of the action. $\square$

Lemma 3 (A separating family of intertwiners). *For each $f \in C_c(X)$, the formula initially defined for $\xi \in C_c(H)$ by*

$$(W_f \xi)(x) = \int_H \xi(h) f(h^{-1}x) dh \quad (2)$$

extends to a bounded operator $W_f : L^2(H) \rightarrow L^2(X, \mu)$ satisfying

$$W_f \lambda(k) = \sigma(k) W_f \quad (k \in H).$$

Moreover,

$$\bigcap_{f \in C_c(X)} \ker W_f = \{0\}.$$

Proof. Set

$$a_f(x) = \int_H |f(h^{-1}x)| dh.$$

Properness implies that a_f is continuous: near a fixed x , the h for which $h^{-1}x \in \text{supp } f$ lie in one compact subset of H , so ordinary compact-support dominated convergence applies. The function is H -invariant and, on the quotient $H \backslash X$, its support is contained in the compact set which is the image of $\text{supp } f$. Consequently $M_f := \sup_X a_f < \infty$.

By Cauchy–Schwarz and Tonelli,

$$\begin{aligned} |W_f \xi(x)|^2 &\leq a_f(x) \int_H |\xi(h)|^2 |f(h^{-1}x)| dh, \\ \|W_f \xi\|_2^2 &\leq M_f \int_H |\xi(h)|^2 \left(\int_X |f(h^{-1}x)| d\mu(x) \right) dh \\ &= M_f \|f\|_{L^1(X, \mu)} \|\xi\|_2^2. \end{aligned}$$

The last equality is exactly the left invariance of the Haar-system measure $\mu = \lambda^u$. Thus W_f extends boundedly. A change of variables $h = k\ell$ gives

$$W_f \lambda(k) \xi(x) = \int_H \xi(\ell) f(\ell^{-1}k^{-1}x) d\ell = \sigma(k) W_f \xi(x).$$

We next record a pointwise fact needed for separation. For every $\xi \in L^2(H)$, $W_f \xi$ has a continuous representative. Indeed, on any compact subset of X , properness confines the supports of $h \mapsto f(h^{-1}x)$ to a common compact subset of H , and the map $x \mapsto [h \mapsto f(h^{-1}x)]$ is continuous into $L^2(H)$. Approximation of ξ by $C_c(H)$ therefore gives locally uniform convergence of the corresponding functions in (2).

Suppose now that $W_f \xi = 0$ in $L^2(X, \mu)$ for every $f \in C_c(X)$. A Haar system has full support, so continuity forces each $W_f \xi$ to vanish pointwise. Fix $x_0 \in X$. Its orbit $O = Hx_0$ is closed, and the map

$$H \longrightarrow O, \quad h \longmapsto h^{-1}x_0$$

is a homeomorphism. Given $\phi \in C_c(H)$, define $g \in C_c(O)$ by $g(h^{-1}x_0) = \phi(h)$. The standard compact-support extension lemma for closed subspaces of locally compact Hausdorff spaces supplies $f \in C_c(X)$ with $f|_O = g$. For clarity, this lemma needs no normality assumption on X : extend g by 0 at infinity from $O \cup \{\infty\}$ in the compact Hausdorff one-point compactification of X , apply Tietze there, and multiply by a compactly supported cutoff equal to 1 on $\text{supp } g$. Then

$$0 = (W_f \xi)(x_0) = \int_H \xi(h) \phi(h) dh.$$

As ϕ was arbitrary and $C_c(H)$ is dense in $L^2(H)$ for the Radon Haar measure, $\xi = 0$. This proves the common-kernel assertion. $\square$

Lemma 4 (Separating intertwiners and coefficients). *Let π and ρ be unitary representations of a locally compact group. If a family of bounded intertwiners $T_i : \mathcal{H}_\pi \rightarrow \mathcal{H}_\rho$ has $\bigcap_i \ker T_i = \{0\}$, then*

$$A_\pi(H) \subseteq A_\rho(H).$$

Proof. The orthogonal complement of $\overline{\text{span}}_i \text{ran } T_i^*$ is $\bigcap_i \ker T_i$, so the former space is all of $\mathcal{H}_\pi$. It is therefore enough to consider coefficient vectors

$$\xi' = \sum_{i=1}^m T_i^* \alpha_i, \quad \eta' = \sum_{j=1}^n T_j^* \beta_j.$$

For every group element k , the adjoint intertwining relation gives

$$\begin{aligned} \langle \pi(k) \xi', \eta' \rangle &= \sum_{i,j} \langle \pi(k) T_i^* \alpha_i, T_j^* \beta_j \rangle \\ &= \sum_{i,j} \langle \rho(k) T_j T_i^* \alpha_i, \beta_j \rangle. \end{aligned}$$

This is a finite sum of coefficients of ρ . Approximating arbitrary $\xi, \eta \in \mathcal{H}_\pi$ by such ξ', η' and using

$$\|\pi_{\xi,\eta} - \pi_{\xi',\eta'}\|_{B(H)} \leq \|\xi - \xi'\| \|\eta\| + \|\xi'\| \|\eta - \eta'\|$$

proves that every coefficient of π belongs to $A_\rho(H)$. □

Proof of Theorem 1. Apply Lemmas 3 and 4 with $\pi = \lambda$ and $\rho = \sigma$. They give directly

$$A(H) = A_\lambda(H) \subseteq A_\sigma(H).$$

Together with the source reduction (1), this gives $A_\sigma(H) = A(H)$ and

$$R_u(A(\mathcal{G})) = A(H).$$

This is surjectivity. Contractivity and multiplicativity were already established in the source at the stated generality. □

5 Scope, checks, and novelty

Why the former extra hypotheses disappear. The argument never restricts λ^u to H and never assigns positive measure to an individual orbit. Thus it applies even when $\lambda^u(H) = 0$, as in nondiscrete pair groupoids. It uses only left Haar measure and left invariance, so nonunimodular isotropy groups cause no modular factor. Every integral is first taken over compact supports; no separability, sigma-finiteness, Borel section, or disintegration theorem is used.

Verification. The proof was audited at its four nonformal points: (i) the shear-map proof of properness; (ii) the Schur–Cauchy bound for W_f ; (iii) continuity plus compact-support extension from a closed orbit; and (iv) the adjoint-range coefficient calculation in Lemma 4. No computational check is relevant. The most valuable human check is a line-by-line comparison of (1) with Claim 1 in the source proof.

Bounded novelty search. On 27 August 2026, bounded searches used the exact arXiv id and title, the authors’ names, and the phrases “Fourier algebra groupoid restriction isotropy surjective”, “ $A(\mathcal{G}) \rightarrow A(\mathcal{G}_u^u)$ ”, and “proper action Koopman regular representation”. The cached source, including its May 2026 update, still contains Remark 3.10. The searches found the source and general proper-action/Koopman background, but no later work stating this countability-free restriction theorem. This is evidence, not a guarantee, of novelty.

Human-review recommendation. Treat this as a likely-valid candidate full solution pending expert review. In particular, verify the compact-support control in Lemma 3, the closed-orbit extension argument, and the source reduction. If those checks pass, the result answers Remark 3.10 exactly.

References

- [1] J. DeGaetani and M. Ghandehari, *On the restriction maps of the Fourier and Fourier-Stieltjes algebras over locally compact groupoids*, arXiv:2411.14624 (2024; updated 2026).
- [2] G. Arsac, *Sur l'espace de Banach engendré par les coefficients d'une représentation unitaire*, Publ. Dép. Math. (Lyon) 13 (1976), no. 2, 1–101.
- [3] E. Kaniuth and A. T. Lau, *Fourier and Fourier–Stieltjes Algebras on Locally Compact Groups*, Mathematical Surveys and Monographs 231, American Mathematical Society, 2018; see in particular Section 2.8.